



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ORGANIC CHEMISTRY VI				
Type of Course	Major/Minor				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Preliminary ideas about common rearrangement reactions, pericyclic reactions and gives an insight to various spectroscopic techniques.				
Course Summary	This course explores common rearrangement reactions, pericyclic reactions and gives an insight to various spectroscopic techniques.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and apply various rearrangement reactions to electron deficient Carbon, Nitrogen and Oxygen	U	C	Test /Seminar
CO2	Comprehend and utilize pericyclic reactions, specifically electrocyclic, cycloaddition and sigmatropic rearrangement reactions	U	p	Dicussion/ Assignment
CO3	Able to explain the principles of UV-Visible-Spectroscopy, IR spectroscopy, interpret their spectra and use Mass spectrometry in molecular mass determination	An	P	Quizes/Test

CO4	Understand and analyze both H NMR and C NMR spectra of simple organic molecules for structure elucidation of organic compounds	Ap	P	Discussion/Seminar /Assignment
CO5	Perform various chemistry practicals including preparation, estimation techniques, column chromatography and identification of unknown molecule via spectroscopic analysis	An	P	Viva Voce/Observation of practical skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Common rearrangement reactions		8	18
	1	Rearrangement to electron deficient Carbon-Wagner - Meerwin rearrangement, Pinacol-Pinacolone rearrangement, Tiffeneau-Demjanov, Dienol-Phenol.	2	
	2	Rearrangement to electron-deficient Nitrogen - Beckmann, Lossen, Hofmann, Curtius.	2	
	3	Rearrangement to electron-deficient Oxygen - Baeyer-Villiger, Dakin reaction.	2	
	4	Acyl carbene rearrangements - Wolff, Arndt-Eistert synthesis.	1	
	5	Anionic rearrangements - Favorskii and Benzilic acid rearrangements.	1	
II	Pericyclic reactions		11	24
	6	Electrocyclic reaction – ring-opening reaction and ring closure reactions in butadiene and hexatriene, con rotation and disrotation of HOMO and LUMO of butadiene and hexatriene for product formation.	2	
	7	Cycloaddition reactions - 2+2 and 4+2 cycloadditions – antarafacial and suprafacial additions, Examples for thermal and photochemical cyclo addition reactions.	2	
	8	Diels alder reaction - dienes and dienophiles in Diels alder reaction, distereoselectivity, hetero Diels-Alder reaction.	2	
	9	Dipolar cycloaddition - Huisgen cycloaddition, Click chemistry (azide-alkyne cycloaddition as example).	1	

	10	Sigmatropic rearrangement - Sigmatropic rearrangements: 1,3 and 1,5 and 1,7 shifts of hydrogen atoms (explanation based on frontier molecular orbitals).	2	
	11	Cope rearrangement and Claisen rearrangement, 2,3-rearrangement, chelotropic reaction, FMO and Moebius-Hückel Approach.	2	
III	UV-Visible-spectroscopy and Mass spectrometry		13	28
	12	UV-Visible-spectroscopy- basic principles, Factors affecting redshift and blueshift, λ_{max} calculation for dienes and α,β -unsaturated carbonyl compounds and polyenes.	3	
	13	IR spectroscopy- basic principles, Factors affecting absorption frequencies, Fingerprint and functional group region.	2	
	14	IR spectra of functional groups-alkenes, Alkynes, Aromatic compounds, Alcohols, Phenols, Carbonyl, Carboxylic acid derivatives, nitro, cyano, sulfoxide.	2	
	15	Mass spectrometry- Theory, Molecular ion peak, Fragment ions, Molecular mass determination, Metastable ion.	3	
	16	Isotopic effect, N Rule, Index of hydrogen deficiency, McLafferty rearrangement, Ionization methods.	3	
IV	NMR Spectroscopy		13	28
	17	NMR Spectroscopy - Basic principles, Chemical shift values in low resolution spectra	3	
	18	High resolution H NMR spectra: Spin-spin splitting, Pascals triangle for Splitting patterns and calculation of coupling constant, Factors affecting coupling constant	3	
	19	Interpretation of ^1H NMR and ^{13}C NMR spectra of simple organic molecules	3	
	20	Structure elucidation of simple organic compounds using UV, IR and ^1H NMR spectroscopic techniques.	4	
V	Practicals		30	
	1.	Introduction to organic lab	4	
	2	1. Estimation of aniline/phenol 2. Estimation of glucose organic compounds by colorimetry 3. Estimation of drug molecules by titration/colorimetry 4. Double stage preparations (Synthesis of dihydroxy tripterycene from anthracene and hydroquinone, reductive amination and its structure analysis) - any one 5. Cannizzaro reaction of p-chlorobenzaldehyde and isolation of products	20	

		6. Column chromatography 7. Identification of unknown molecule via spectroscopic analysis (measure the spectra and analyse if the instruments are accessible, otherwise analyse the provided the spectra)		
	3	Open ended	6	

References

1. Organic Chemistry, by Jonathan Clayden, Nick Greeves, Stuart Warren, Oxford University Press
2. March's Advanced Organic Chemistry: Reactions, Mechanisms and Structure, [Michael B. Smith](#), Wiley
3. Organic Chemistry Paperback, Francis Carey, McGraw-Hill Education
4. Advanced Organic Chemistry, David E. Lewis, Oxford Univ Pr; Illustrated edition
5. Principles of Organic Synthesis, R. O. C. Norman & J. M. Coxon, 3rd Ed., CRC press, 2000.
6. Organic Mechanisms- Reactions, Stereochemistry and Synthesis, R. Bruckner, Springer 2007.
7. The Art of Writing Reasonable Reaction Mechanisms, R. B. Grossman, 3rd Ed., Springer, 2019.
8. Name reactions J J Li, Springer.
9. Spectrometric Identification of Organic Compounds, 8ed Paperback – 1 March 2022
10. Robert M. Silverstein, Francis X. Webster, David J. Kiemle, Wiley.
11. Introduction to Spectroscopy, D. L. Pavia, 5th Ed., Cengage, 2015.
12. Organic spectroscopy, William Kemp. MACMILLAN; SECOND edition
13. Spectroscopy of Organic compounds, P S Kalsi. New Age International Private Limited
14. Vogel's practical organic chemistry. Pearson India
15. Quinone Synthesis and a Visual Introduction to Column Chromatography: An Undergraduate Experiment, Danielle L. Pearson and Russell R. A. Kitson, J. Chem. Educ. 2022, 99, 3731–3734, <https://doi.org/10.1021/acs.jchemed.1c00940>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	-	-	-	-	-	3		2	1	3		2
CO 2	2		-	-	-	-	3		2	1	3		2
CO 3	-	-		-	2	2	3		2	2	3		3
CO 4	-	-			3	3	3		2	2	3		3

CO 5	-		3	-	3	3	3		3	2	3		3
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/viva/seminar	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2		✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5	✓	✓		✓